

Product Overview

Protpure DEAE Agarose resins are weak anion exchangers based on a 6% highly cross-linked agarose matrix.

They are designed for high-resolution purification of proteins and other biomolecules, offering excellent chemical stability, reproducibility, and scalability for process development and manufacturing.

- High Binding Capacity
Consistent & Reproducible
- Robust Chemical Stability
Wide pH & CIP Range
- Scalable Performance
From Lab to Manufacturing
- Agarose-Based
Bioprocess Platform

Comparative Performance Overview

Parameter	DEAE Agarose Fast Flow	DEAE Agarose Precise	DEAE Agarose HR	Typical Market Specification
BioProcess Resin	Yes	Yes	Yes	Yes
Ion Exchanger Type	Weak anion exchanger	Weak anion exchanger	Weak anion exchanger	Weak anion exchanger
Ionic Capacity	0.12–0.15 mmol (Cl ⁻)/mL resin	0.12–0.15 mmol (Cl ⁻)/mL resin	0.12–0.15 mmol (Cl ⁻)/mL resin	0.88–1.08 mmol (Cl ⁻)/mL resin or /mg dry material
Ligand	Diethylaminoethyl	Diethylaminoethyl	Diethylaminoethyl	Diethylaminoethyl
Matrix	6% spherical cross-linked agarose	6% spherical cross-linked agarose	6% spherical cross-linked agarose	6% cross-linked agarose, spherical
Particle Size Range	45–165 µm	45–105 µm	25–55 µm	45–165 µm
Particle Size (d50V)	~90 µm	~65 µm	~35 µm	90 µm
Dynamic Binding Capacity	≈90 mg BSA/mL resin	≈110 mg BSA/mL resin	≈120 mg BSA/mL resin	40–100 mg BSA/mL resin or g/L bed vol
Pressure/Flow Specification	250–450 cm/h, 0.1 MPa, 15 cm bed height	100–300 cm/h, 0.1 MPa, 15 cm bed height	80–150 cm/h, 0.1 MPa, 15 cm bed height	300–600 cm/h, 0.1 MPa, 10 cm bed height, 35 cm/h, 0.069 MPa (dry to wet bed)
pH Stability (CIP)	2–14	2–14	2–14	2–14
pH Stability (Operational)	2–12	2–12	2–12	2–12
Chemical Stability	Stable in 1.0 M NaOH, 8 M urea, 6 M guanidine hydrochloride, 70% ethanol	Stable in 1.0 M NaOH, 8 M urea, 6 M guanidine hydrochloride, 70% ethanol	Stable in 1.0 M NaOH, 8 M urea, 6 M guanidine hydrochloride, 70% ethanol	Stable in 1.0 M NaOH, 8 M urea, 6 M guanidine hydrochloride, 70% ethanol
Storage Conditions	4–30°C, 20% Ethanol	4–30°C, 20% Ethanol	4–30°C, 20% Ethanol	4–30°C, 20% Ethanol

Applications

- Protein purification and intermediate polishing
- Enzyme and peptide purification
- Vaccine and recombinant protein downstream processing
- Separation of biomolecules with weak anionic nature

Key Benefits

- Optimized weak anion exchange for high selectivity and yield
- Excellent flow properties for faster process development
- Robust across a wide range of pH and cleaning conditions
- Reliable performance with lot-to-lot consistency
- Suitable for scale-up from R&D to commercial manufacturing



Evaluation samples and application support available upon request.
Contact us to discuss your process requirements.